SPEC 5850
April, 2011

SunGen® XLPE, Photovoltaic Wire 2000 V, UL Type PV/RHH or RHW-2 or 600 V USE-2 or c(UL) RWU90, 1 kV, 90°C



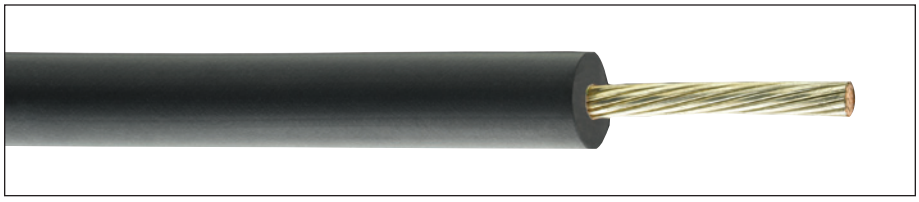
Product Construction:

Conductor:
- 18 AWG thru 2 AWG tinned coated compressed copper. Class C stranding per ASTM B33 and B8
- 1 AWG thru 1000 kcmil tinned coated compressed copper. Class B stranding per ASTM B33 and B8

Insulation:
- Flame-retardant Cross-Linked Polyethylene (XLPE), black

Print:
- 18 AWG and 16 AWG: GENERAL CABLE® (PLANT OF MFG) SUNGEN® 2000 V PV WIRE (SIZE) 90°C WET OR DRY SUN RES (UL) -40°C VW-1 MONTH/YEAR OF MFG SEQUENTIAL FOOTAGE MARK
- 14 AWG thru 1000 kcmil: GENERAL CABLE® (PLANT OF MFG) SUNGEN® 2000 V PV WIRE OR RHH OR RHW-2 OR 600 V USE-2 (SIZE) 90°C WET OR DRY SUN RES (UL) -40°C VW-1 c(UL) RWU90 1000 V DIR BUR MONTH/YEAR OF MFG SEQUENTIAL FOOTAGE MARK

Options:
- Bare copper conductor
- Other stranding options are available upon request
- Colors available on request



Applications:
- Single conductor, sunlight-resistant, photovoltaic wire rated 90°C wet or dry, 2000 V for interconnection wiring of grounded and ungrounded photovoltaic power systems as described in Section 690.31(A) and other applicable parts of the National Electrical Code (NEC), NFPA 70

Features:
- Rated 90°C wet and dry
- Deformation-resistant at high temperatures
- Excellent moisture resistance, exceeds UL 44
- Stable electrical properties over a broad temperature range
- Excellent resistance to crush and compression cuts
- Resistant to most oils and chemicals
- UV/sunlight-resistant
- Meets cold bend test at -40°C

Compliances:
Industry Compliances:
- UL 4703 Type PV, UL File # E323451
- National Electrical Code (NEC)
- ICEA S-95-658/NEMA WC70
- UL 44 Type RHH or RHW-2, UL File # E90494
- UL 854 Type USE-2 for 600 V, UL File # E90499
- c(UL) RWU90 1 kV, UL File # E90494

Flame Test Compliances:
- UL 1581 VW-1
- For sizes 1/0 and larger: IEEE 383, IEEE 1202/CSA FT4

Other Compliances:
- EPA 40 CFR, Part 261 for leachable lead content per TCLP
- OSHA Acceptable

Packaging:
- Material cut to length and shipped on non-returnable wood reels

CATALOG NUMBER	COND. SIZE (AWG/ kcmil)	COND. STRAND	NOMINAL COND. O.D.		MINIMUM AVG. INSULATION THICKNESS		NOMINAL CABLE DIAMETER		COPPER WEIGHT		NET WEIGHT	
			INCHES	mm	INCHES	mm	INCHES	mm	LBS/ 1000 FT	kg/km	LBS/ 1000 FT	kg/km
18 AWG - 1000 kcmil CONDUCTORS												
12421.711800*	18	19/.0092	0.045	1.14	0.075	1.90	0.199	5.05	5	7	16	24
12421.711600*	16	19/.0117	0.056	1.42	0.075	1.90	0.210	5.33	8	12	21	31
12421.711400	14	19/.0142	0.070	1.78	0.075	1.90	0.224	5.69	13	19	32	48
12421.711200	12	19/.0185	0.088	2.23	0.075	1.90	0.244	6.20	20	30	42	62
12421.711100	10	19/.0234	0.112	2.84	0.075	1.90	0.268	6.81	32	48	57	85
12421.710800*	8	19/.0295	0.143	3.63	0.085	2.16	0.326	8.28	50	74	87	129
12421.710600*	6	19/.0372	0.184	4.67	0.085	2.16	0.363	9.22	81	121	123	183
12421.710400*	4	19/.0469	0.234	5.94	0.085	2.16	0.406	10.31	129	192	181	269
12421.710200*	2	19/.0526	0.296	7.52	0.085	2.16	0.474	12.04	205	305	266	396
12421.710100*	1	19/.0664	0.323	8.20	0.105	2.67	0.538	13.66	258	384	350	521
12421.715100*	1/0	19/.0740	0.370	9.40	0.105	2.67	0.586	14.88	326	485	429	638
12421.715200*	2/0	19/.0837	0.410	10.41	0.105	2.67	0.631	16.03	411	611	527	784
12421.715300*	3/0	19/.0940	0.460	11.68	0.105	2.67	0.674	17.12	518	771	647	963
12421.715400*	4/0	19/.1055	0.520	13.21	0.105	2.67	0.737	18.72	653	972	796	1184
12421.716250*	250	37/.0822	0.558	14.17	0.120	3.05	0.804	20.42	772	1149	938	1396
12421.716300*	300	37/.0900	0.611	15.52	0.120	3.05	0.857	21.77	926	1378	1106	1646
12421.716350*	350	37/.0972	0.661	16.79	0.120	3.05	0.907	23.04	1063	1582	1257	1870
12421.716400*	400	37/.1040	0.706	17.93	0.120	3.05	0.952	24.18	1235	1838	1441	2144
12421.716500*	500	37/.1159	0.789	20.04	0.120	3.05	1.035	26.29	1509	2246	1737	2585
12421.716600*	600	61/.0992	0.866	22.00	0.135	3.43	1.142	29.01	1883	2802	2157	3211
12421.716750*	750	61/.1109	0.968	24.59	0.135	3.43	1.244	31.60	2316	3447	2620	3900
12421.717000*	1000	61/.1280	1.117	28.37	0.135	3.43	1.393	35.38	3088	4595	3437	5115

Dimensions and weights are nominal; subject to industry tolerances.
* Non-stock item; minimum runs apply. Please contact Customer Service for price and delivery.

General Cable
Phone: 888-593-3355
www.generalcable.com

c UL LISTED SunGen